

Final Project

1. Project Overview

The final project is for you to apply what you have learned in this class to a problem in computer vision that interests you. You can either choose one of the suggested project topics below (Option 1) or come up with your own idea (Option 2).

Your project must investigate a **scientific question**. It is not enough to do a literature review, nor to directly reimplement a method without making any changes to it. A good project is one where you have scoped out a reasonable problem and demonstrated first experimental evidence that your proposal is viable. We expect at least 15 hours of work per team member on the project, and we are allocating time in the schedule to accommodate this (the workload will be consistent with problem set expectations during the equivalent time period). We expect four-person projects to be about 4/3 the amount of work and content as three-person projects.

Further below you will find instructions regarding the project proposal, presentation, and report, and detailed grading rubrics.

2. Project Topic

Option 1: Choose one of the suggested project topics:

3D Gaussian Splatting for Scene Reconstruction

3D Gaussian splatting[1] is an emerging technique for scene reconstruction that represents environments as a collection of anisotropic Gaussian primitives, allowing for a continuous and differentiable rendering process. This project focuses on implementing and evaluating this approach to reconstruct dense, photorealistic 3D scenes from sparse multi-view image data. Unlike mesh-based rendering methods, Gaussian splatting enables flexible manipulation of transparency, size, and color attributes, making it particularly suitable for complex lighting and texture scenarios. By leveraging multi-view datasets such as NeRF[2] or DTU, the project aims to reconstruct environments with high fidelity while optimizing rendering speed for real-time applications.

The project will start with pre-processing data, including camera pose estimation using tools like COLMAP. Gaussian primitives will then be initialized based on 3D point clouds derived from multi-view images. A differentiable rendering pipeline will optimize the parameters of the Gaussians to minimize reconstruction error compared to the ground truth images. The results will be evaluated using metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), with visual comparisons against traditional reconstruction methods. This project could be particularly impactful for applications like virtual reality, immersive gaming, and

architectural visualization, as it offers a seamless blend of computational efficiency and photorealistic accuracy.

References

- [1] Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Trans. Graph.*, 42(4), 139-1.
- [2] Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2021). Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1), 99-106.

Fine-Tuning Large Language Models for Domain-Specific Applications

Large Language Models (LLMs) such as GPT[1] and BERT[2] have demonstrated remarkable performance across a variety of natural language processing tasks. However, general-purpose models often lack the specificity required for niche domains like legal document analysis, medical text summarization, or scientific literature interpretation. This project will focus on fine-tuning an LLM to perform a domain-specific task, such as summarizing clinical reports, identifying legal precedents, or generating scientific abstracts. Fine-tuning involves adapting a pre-trained model to a smaller dataset from the target domain, enabling it to specialize while retaining the general language understanding developed during pre-training.

The project will begin with curating a domain-specific dataset, such as PubMed articles for medical text or Caselaw Access Project documents for legal analysis. Data preprocessing will include tokenization, cleaning, and splitting into training and validation sets. Fine-tuning methods like LoRA (Low-Rank Adaptation) or Adapter Layers will be employed to efficiently adapt the model while minimizing computational costs. The fine-tuned model will be evaluated using task-specific metrics such as BLEU, ROUGE, or F1 scores, alongside qualitative analyses like error inspection. This project is particularly relevant for improving task efficiency in specialized fields, enhancing interpretability in high-stakes domains, and developing customized AI tools for specific professional needs.

References

- [1] Schulman, J., Zoph, B., Kim, C., Hilton, J., Menick, J., Weng, J., Uribe, J. F. C., Fedus, L., Metz, L., Pokorny, M., Gontijo-Lopes, R., Zhao, S., Vijayvergiya, A., Sigler, E., Perelman, A., Voss, C., Heaton, M., Parish, J., Cummings, D., Nayak, R., Balcom, V., Schnurr, D., Kaftan, T., Hallacy, C., Turley, N., Deutsch, N., Goel, V., Ward, J., Konstantinidis, A., Zaremba, W., Ouyang, L., Bogndonoff, L., Gross, J.,

Medina, D., Yoo, S., Lee, T., Lowe, R., Mossing, D., Huizinga, J., Jiang, R., Wainwright, C., Almeida, D., Lin, S., Zhang, M., Xiao, K., Slama, K., Bills, S., Gray, A., Leike, J., Pachocki, J., Tillet, P., Jain, S., Brockman, G., and Ryder, N. ChatGPT: Optimizing Language Models for Dialogue, November 2022. URL <https://openai.com/blog/chatgpt/>.

[2] Devlin, J. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.

Generative Adversarial Networks (GANs) for Artistic Image Creation

Generative Adversarial Networks (GANs)[1] have become a cornerstone of generative modeling, capable of producing highly realistic and creative outputs. This project will explore the use of GANs to generate artistic images that mimic the styles of famous painters or create entirely new artistic expressions. By training a GAN on datasets such as WikiArt, the model will learn to synthesize images that reflect the aesthetic and compositional qualities of specific artistic styles. The project will address challenges such as balancing creative diversity with output fidelity and ensuring training stability.

The implementation will involve advanced GAN architectures like StyleGAN2, which allows for fine-grained control over the generated images' attributes. The dataset will be preprocessed to ensure uniform resolution and style representation. Training stability will be enhanced through techniques such as progressive growing and spectral normalization. The project's output will be evaluated using both quantitative metrics like Fréchet Inception Distance (FID) and qualitative assessments through user studies. This project has potential applications in digital art creation, virtual environments, and tools for assisting artists in their creative workflows. By combining technical innovation with artistic exploration, the project aims to push the boundaries of AI-driven creativity.

References

[1] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). Generative adversarial nets. *Advances in neural information processing systems*, 27.

Computer-Aided Diagnosis

Healthcare is being transformed by advances in computer vision algorithms. Radiology departments across the world have been successfully piloting software that automatically identifies and characterizes disease in medical images. This allows physicians to prioritize patients with severe disease, and also helps identify incidental findings (disease that is discovered unintentionally, during evaluation/treatment for a different disease). Convolutional neural networks are already competitive with radiologists at detecting a number of common diseases such as pneumonia in chest

x-ray [2] and hemorrhage on CT [1]. Still, many challenges remain for making such algorithms reliable diagnostic tools:

1. Incorporating textual information (previous clinical reports of the patient), longitudinal imaging (previous scans of the patient), multiple imaging modalities (multiple MRI sequences), or other patient data (demographics, lab values, genetics)
2. Learning to classify rare diseases (transfer learning)
3. Generalizing between clinical sites and scanners (robustness)
4. Interpretability and explainability (some of the techniques from Miniplaces part 1 are relevant here)

There are a number of large public datasets for different diseases/image types The Cancer Imaging Archive <https://www.cancerimagingarchive.net>, MIMIC-CXR <https://physionet.org/content/mimiccxr/2.0.0/>, DeepLesion <https://nihcc.app.box.com/v/DeepLesion>, etc.), as well as many challenges that focus on medical image analysis (<https://grand-challenge.org/challenges/>, Medical Segmentation Decathlon <http://medicaldecathlon.com/>). Choose a disease or challenge that interests you and build a network that tries to further the state of the art.

References

- [1] Monika Grewal et al. “RADNET: Radiologist Level Accuracy using Deep Learning for HEMORRHAGE detection in CT Scans”. In: CoRR abs/1710.04934 (2017). arXiv: 1710 . 04934. url: <http://arxiv.org/abs/1710.04934>.
- [2] Pranav Rajpurkar et al. “CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning”. In: CoRR abs/1711.05225 (2017). arXiv: 1711.05225. url: <http://arxiv.org/abs/1711.05225>.

Campus Navigation

At some point, we’ve all gotten lost in the Campus. It can be a difficult building to navigate if you’ve never seen it before. But once you know your way around, it’s not too hard. It seems, then, that we should be able to automate the complicated process of giving people directions in our campus. This project is to build a system that will guide visitors around our school. The user could use their phone to give a destination (just like in Google Maps) and then take a picture of where they are. The system will recognize the location and orientation in space and will tell the user in which direction to continue. The user can take pictures along the way to verify the path and get continued directions. For your first implementation, you can also build a web system; it doesn’t necessarily need to be phonebased. You may want to gather some “training” images from around our school. How can you account for the differences between the

training images and the images that users will take? From an image, how can you determine which direction to go in? Do you need to localize the user exactly to give directions? Can you take a hierarchical approach (it may first help to know what floor the user is on)?

3D Printing Martian Rocks

Rover missions have made visualizing Mars's terrain a reality. With the touchdown of Curiosity on Mars, we have had a first person view of craters, mountains and ancient rivers with amazing definition. But what if we view Mars in 3D in person rather than 2-Dimensionally? We can, and the answer lies in utilizing the power of stereo-vision. Stereo reconstruction algorithms try to solve this question. Given two images, a stereo reconstruction algorithm aims to recover the full 3D structure of the images. By matching two color channels (red and blue) of features of the stereo image and utilizing a function which relates the difference in location of features in the two color channels, we can recover the distance between the rover and the selected features of Martian terrain. With this we can interpolate the distance in order to create a continuous depth-map. From this depth-map, we can transform the stereo image into a 3D object. We show an example image from Martian rover Curiosity's journey in figure 1.

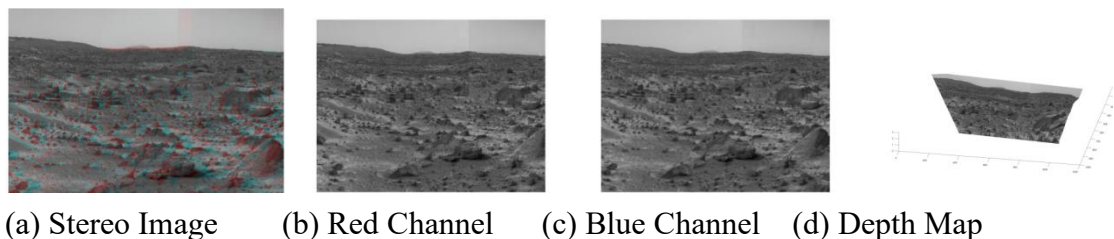


Figure 1: Preliminary Stereo Reconstruction Example

In this project, we aim to develop an algorithm that can take stereo images from Curiosity's Mars Mission 1 and past archives of older rover missions 2 and transform them into 3D models for printing. Even with stereo reconstruction, we would still have to solve for the hidden surfaces such as the side of the rock the viewer cannot see. This would be solved initially with a statistical model that would fill-in the unseen holes of 3D rocks/scenes. If this statistical model proves rudimentary, we would like to take a step further and implement a 3D-GAN which would create a plausible model of the 3D terrain as described in previous works 3.

1<https://mars.nasa.gov/mars3d/>

2<https://mars.nasa.gov/MPF/mpf/stereo-arc.html>

3<https://papers.nips.cc/paper/6096-learning-a-probabilistic-latent-space-of-object-shapes-via-3d-generative-adversarial-modeling.pdf>

3D Shape Reconstruction

In recent years we have witnessed an explosion in the amount of 3D data that we can generate and store. On one hand, better 3D modeling tools have enabled designers to build 3D models easily, resulting in an expansion in the size of 3D CAD model repositories. On the other hand, commodity depth sensors have allowed ordinary

people to conveniently capture their own 3D scans. We need good techniques for leveraging such 3D content, in order to design algorithms that successfully understand our 3D world. However, due to fundamental challenges in dealing with 3D representations and processing, there are still many open research issues. Two key research problems are: (1) 3D shape reconstruction based on a single image, and (2) shape part level segmentation. Existing algorithms are usually evaluated on small datasets with a few hundreds of models, even though millions of 3D models are now available on the Internet. Thanks to the efforts of the ShapeNet [1], we can now use a much larger and varied repository of 3D models to develop and evaluate new algorithms in computer vision and computer graphics. In this track, we focus on the first problem aiming to evaluate the performance of 3D reconstruction based on single image on a subset of the ShapeNet dataset.

In the 3D reconstruction challenge, the task is defined as reconstructing a 3D shape, given a single image as input. We use the ShapeNetCore subset of ShapeNet which contains about 48,600 3D models over 55 common categories. There is a 70%/10%/20% training/validation/test split from this dataset and voxels are used as the output 3D representations. Datasets and evaluation code are available at <https://shapenet.cs.stanford.edu/iccv17/>.

References

[1] A. X. Chang, T. Funkhouser, L. Guibas, P. Hanrahan, Q. Huang, Z. Li, S. Savarese, M. Savva, S. Song, H. Su, et al., “Shapenet: an information-rich 3d model repository,” arXiv preprint arXiv:1512.03012, 2015.

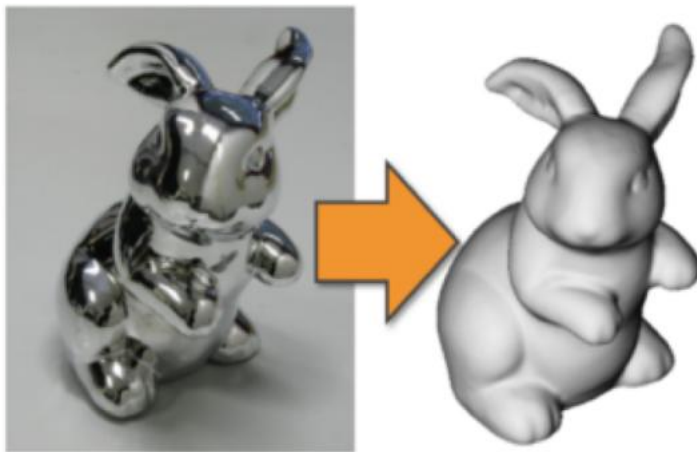


Figure 1: Converting a single 2D image into a 3D shape

Reconstructing 3D Structure from Drawing & Sketches

As reconstructing 3D information from multiple images are actively explored in computer vision and graphics, one came to wonder what we could do to data from pre-photograph era: sketches and drawings. Specifically, Field Lane, a notorious slum in early 1800s London, where Charles Dickens used as part of the settings of *Oliver Twist*.

You can find various early 1800s sketches about Field lane on the internet. The task is to find correspondences between sketches and reconstruct a reasonable 3D Filed Lane

using those sketches, where one may explore the virtual street of London in early 1800s. A introductory video can be found here [1].

References

- [1] Reconstructing field lane. <https://www.dropbox.com/s/tmk4oufi433n5eq/ImagingDickensLondon.mp4?dl=0>.

Scene Segmentation

Scene segmentation, or pixel-wise dense labeling, is one of the key problems in computer vision. Specifically, one aims to give every pixel a label for images, which is more fine-grained than image-level recognition or object detection tasks. Existing dataset like ADE20K [1] has a large collection of carefully labeled natural scene images, with over 3000 classes. And there are also existing works on image segmentation tasks, one of the first is Fully Convolutional Neural Networks (FCN) [2]. In this project, we plan to take part of the annotations, say the most common 150 objects in the ADE20K dataset, to train our segmentation model. A good reference for state-of-the-art performance can be found on the Scene Parsing Challenge website, <http://sceneparsing.csail.mit.edu/>. We do not aim for the best performance, but we want the project to be exploratory where an original model can be proposed (e.g. instance segmentation, improving semantic edges, compress large models).

References

- [1] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, “Semantic understanding of scenes through the ade20k dataset,” arXiv preprint arXiv:1608.05442, 2016.
- [2] J. Long, E. Shelhamer, and T. Darrell, “Fully convolutional networks for semantic segmentation,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 3431–3440.

Visual Question Answering (VQA) in a Toy World

An old saying goes “1 image worths 1000 words”. In matter of facts, so many questions can be answered with the information provided by one single image. This task, answering questions from images, is usually referred as Visual Question Answering and has been very popular in computer vision during the last years. Given an image, VQA systems try to answer an input question. This problem can be formulated as a multiple-choice problem or an open-ended problem. Since the publication of the VQA dataset [1], many models have been developed to improve the quality of the answer and provide more flexibility. Although many advances have been made in the field, it is a hard problem with so many incontrollable variables. For this reason, we propose a reduced version of the problem, instead of doing VQA on natural images, we want to perform the task in a toy world where the questions and answer refers to elements in this toy world. To further simplify the problem, we want the questions to be multiple-choice instead of open-ended. Luckily, [1] contains a subset of the dataset with this properties.

In this project, we want to develop a model to take part in the VQA-Challenge 1 for abstract scenes in multiple-choice answer. Some other models tried to approach the

same problem [2, 3]; we do not aim for the best performance in the challenge, we want the project to be an exploratory project where an original model is proposed, tested in the dataset and the results are reported.

References

- [1] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. L. Zitnick, and D. Parikh, “Vqa: Visual question answering,” in International Conference on Computer Vision (ICCV), 2015.
- [2] P. Zhang, Y. Goyal, D. Summers-Stay, D. Batra, and D. Parikh, “Yin and yang: Balancing and answering binary visual questions,” ArXiv preprint arXiv:1511.05099, 2015.
- [3] D. Teney, L. Liu, and A. v. d. Hengel, “Graph-structured representations for visual question answering,” ArXiv preprint arXiv:1609.05600, 2016.

NeRF: Neural Radiance Fields

In this project, we are interested in novel view synthesis: synthesizing how a scene looks like from viewpoints that has not been observed before. Specifically, given a set of images of a scene and their associated camera parameters, the goal is to generate high-fidelity images of the scene for arbitrary* viewpoints of interest.

NeRF use classic rendering techniques [2] to render the color of a pixel. To optimize the radiance field, NeRF minimizes the squared loss between the ground truth pixels and the corresponding rendered pixels. One potential project direction is to implement a NeRF from scratch. There are several implementations online [3, 4] which will be helpful references. Another direction is to make these implementations faster. A core limitation of NeRF is its rendering speed and improving constant factors in the code could make a huge impact. Other project directions include addressing or improving a limitation of NeRF. One limitation is that NeRF assumes training images have known associated camera parameters. One project direction is to devise a method to train NeRF with noisy, or even unknown, camera parameters [5]. Another limitation is that NeRF is unable to fit to changing scenes and assumes that the training images are of the same scene under the same snapshot in time. One project direction is to train a NeRF on a scene where the training images vary in lighting conditions or times of day [6]. Furthermore, an additional limitation is that NeRF rendering times are slow; it can take several seconds to render a high resolution view. One project direction is to improve the rendering times of NeRF [7, 8, 9]. Besides addressing the limitations of NeRF, another possible direction is to extend NeRF to other domains. For instance, exploiting NeRF to encode the semantics of the scene [10] or combining NeRF with other representations [11]. This is an exciting field and there are a lot things worth exploring, so feel free to pick one you are interested in. Feel free to refer to this survey [12] for a more comprehensive view of potential improvements, extensions, and followup work to NeRF.

References

- [1] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In ECCV, 2020. 1

- [2] Thomas Porter and Tom Duff. Compositing digital images. In Proceedings of the 11th Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH '84, page 253–259, New York, NY, USA, 1984. Association for Computing Machinery. 1
- [3] Lin Yen-Chen. PyTorchNeRF: a PyTorch implementation of NeRF, 2020. 1
- [4] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Neural radiance fields, 2020. 1
- [5] Zirui Wang, Shangzhe Wu, Weidi Xie, Min Chen, and Victor Adrian Prisacariu. Nerf-: Neural radiance fields without known camera parameters, 2021. 2
- [6] Ricardo Martin-Brualla, Noha Radwan, Mehdi S. M. Sajjadi, Jonathan T. Barron, Alexey Dosovitskiy, and Daniel Duckworth. NeRF in the Wild: Neural Radiance Fields for Unconstrained Photo Collections. In CVPR, 2021. 2
- [7] Alex Yu, Ruilong Li, Matthew Tancik, Hao Li, Ren Ng, and Angjoo Kanazawa. Plenotrees for real-time rendering of neural radiance fields. In arXiv, 2021. 2
- [8] Stephan J. Garbin, Marek Kowalski, Matthew Johnson, Jamie Shotton, and Julien Valentin. Fastnerf: High-fidelity neural rendering at 200fps, 2021. 2
- [9] Peter Hedman, Pratul P. Srinivasan, Ben Mildenhall, Jonathan T. Barron, and Paul Debevec. Baking neural radiance fields for real-time view synthesis, 2021. 2
- [10] Shuaifeng Zhi, Tristan Laidlow, Stefan Leutenegger, and Andrew J Davison. In-place scene labelling and understanding with implicit scene representation. arXiv, 2021. 2
- [11] Suttisak Wizadwongsa, Pakkapon Phongthawee, Jiraphon Yenphraphai, and Supasorn Suwajanakorn. Nex: Real-time view synthesis with neural basis expansion. In CVPR, 2021. 2
- [12] Frank Dellaert and Lin Yen-Chen. Neural volume rendering: Nerf and beyond, 2021. 2

Motion Magnification for Planets

The planets are often called "wandering stars" because they move relative to the background stars. In fact, "wandering stars" translates to "planētes asters" in ancient Greek, from which the word "planet" is derived. This relative motion is usually measured over the course of weeks or months. Using motion magnification, however, you should be able to show that the planets move differently than the stars even over the course of 1/2 hour or 1 hour within one night. Take a sequence of photographs over time of the night sky with stars and one or more planets in view. A reasonable dataset will contain 200 photographs, taken 10 seconds apart. Ideally, use a camera on a tripod with a star-tracking mount. Finally, process the resulting images in a way that will exaggerate the motion of the planet(s) relative to the stars within the field of view. It would also be interesting to compare this to existing phot datasets taken over weeks or months.

Video Super Resolution

Recent work has approached image superresolution using a variety of approaches [1, 2]. Other work has proposed novel loss functions for improved results [3]. How can

we use these findings to do video superresolution? A simple baseline approach is to simply apply image superresolution independently to each frame. However, this is unlikely to result in a smooth video. Consecutive frames are inherently dependent on each other, and we would like to take advantage of this. In this project, you will investigate how we can intelligently apply image superresolution techniques to video. Rather than applying superresolution to each frame independently, you might want to add some sort of temporal smoothing across frames, or incorporate additional constraints into the loss function. Fortunately, procuring high quality training data will not be difficult for this task. You may simply downsample videos of your choice to use for training. We would like this to be an exploratory project in which you propose an original model and/or loss function to improve upon the baseline.

References

- [1] W. Shi, J. Caballero, F. Huszár, J. Totz, A. P. Aitken, R. Bishop, D. Rueckert, and Z. Wang, “Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 1874–1883.
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- [3] J. Johnson, A. Alahi, and L. Fei-Fei, “Perceptual losses for real-time style transfer and superresolution,” in European Conference on Computer Vision, Springer, 2016, pp. 694–711.

Visually Indicated Sounds

When you see photo of a beach, it’s easy to imagine the sound of wind and waves that would naturally accompany such a scene. This ability to correlate sensory experience across modalities is considered to be crucial to the development of perceptual systems that can reason coherently about all the different types of information they take in [3]. Being able to predict sounds from images or silent videos also makes for really fun applications. For example, you could add sounds to a movie from the silent films era, or you could add a soundtrack to a photo album of you on vacation. Recently there has been quite a bit of interest in the computer vision community on several versions of this problem [1, 2, 4]. In this project, your goal will be to take an image or video as input, and predict a sound as output. For training data, you could use any source of videos with audio tracks, where you simply split the video into the visual frames as the inputs to your system and the audio as the output. To represent sounds, you may want to consider the cochleograms used in [1], or you could use a spectrogram representation. Some good datasets to consider are:

1. Drumsticks hitting objects: <http://andrewowens.com/vis/>
2. Talking people: <https://www.robots.ox.ac.uk/vgg/data/voxceleb/>
3. Scenes with audio: <https://www.robots.ox.ac.uk/vgg/data/vggsound/>

As a twist, you could instead try predicting images from sounds. Given audio of birdsong, can you predict what the bird might look like? Feel free to be creative and come up with your own variants on the problem!

References

- [1] A. Owens, P. Isola, J. McDermott, A. Torralba, E. H. Adelson, and W. T. Freeman. Visually indicated sounds. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 2405–2413, 2016. [
- 2] A. Owens, J. Wu, J. H. McDermott, W. T. Freeman, and A. Torralba. Ambient sound provides supervision for visual learning. In European conference on computer vision, pages 801–816. Springer, 2016.
- [3] L. Smith and M. Gasser. The development of embodied cognition: Six lessons from babies. *Artificial life*, 11(1-2):13–29, 2005.
- [4] H. Zhao, C. Gan, A. Rouditchenko, C. Vondrick, J. McDermott, and A. Torralba. The sound of pixels. In Proceedings of the European conference on computer vision (ECCV), pages 570–586, 2018.

Option 2: Come up with your own project proposal:

You could select a topic in computer vision that interests you and create your own project around it. A potential project could focus on an application or on creating new models or improving existing ones:

- Application: You could apply computer vision techniques to a specific application with your background and interest.
- Models: You could create new models or improve previous models or methods, then evaluate them systematically on standard image datasets to demonstrate their strengths and weaknesses.

3. Project Proposal Requirements

The proposal requires you to define the research space you'll investigate, and to describe the research you plan to do within that space. You will: identify a research gap; move from that problem/opportunity to a solution/goal; and come up with an approach for how that might be done. We will read your proposal for content, and you will receive more detailed feedback from your communication instructor and your peers to guide you in the revision and development process. The proposal is due **Friday, Oct. 7 at 23:59 pm** and must be submitted in **PDF format** to iSpace.

If you choose **Option 1** (going with a suggested topic), then your team's project proposal is not required. If you choose **Option 2** (coming up with your own project idea), your team's proposal should answer the following questions. While we will not grade your proposal, we will provide feedback, in particular regarding the scope of your proposed project.

- What is the problem/question that you will be investigating?
- Why is the problem interesting/important?
- What are the most relevant readings (2-4 papers)?
- What data will you use?
- What are the existing methods? Are their implementations available?
- What method or algorithm will you use, and why? What motivates your choice of this approach?
- What computing resources will you use to train and run your model(s)?
- How will you evaluate your results?
- Qualitatively, what kind of results do you expect (e.g. plots or figures)?
- Quantitatively, what kind of analysis (performance metrics, statistical tests etc.) will you use to evaluate and/or compare your results?
- If your approach is successful, what difference will this contribution make?

Formatting: The formatting of the proposal is up to you: it can take the form of sections separated by headings, each addressing one or several of the questions above, or it can be in a narrative format (i.e. without section headings). Whichever option you choose, your proposal should be **written in full sentences, rather than bulleted lists**. The **length** of the proposal should be **1 single-spaced page, or 2 double-spaced pages**, maximum.

Grading: Your proposal will be graded with a check/check plus/check minus.

4. Project Report Requirements

The report should ideally be 5 pages in length (excluding references). We will deduct points for excessively and unnecessarily long reports (a well-written concise report is better than a long and wordy one!). Furthermore, you should write the report together but you must include a section that **lists the individual contributions of each team member**. We will apply a penalty if this section is missing. The report should be structured like a research paper, starting with an abstract, followed by sections for Introduction, Related Work, Methodology, Experimental Results, Discussion, Conclusion, and ending with references. Some of the sections can be combined if you want (specifically, Introduction/Motivation & Related Work as well as Results & Discussion). We recommend you write the report in the [CVPR format](#).

You should describe and evaluate what you did in your project, which may not necessarily be what you hoped to do originally. A small result described and evaluated well will earn more credit than an ambitious result where no aspect was done well. Be accurate in describing the problem you tried to solve. Explain in detail your approach, and specify any simplifications or assumptions you have made. Also demonstrate the limitations of your approach. When doesn't it work? Why? What

steps would you have taken had you continued working on it? Make sure to add references to all related work you reviewed or used.

5. Project Presentation Video

The presentation gives you a chance to provide us with a concise overview of the current state of your project and get us excited to read your report! The presentation should highlight the key contributions your project is making and the insights you've gained while tackling a new research problem. Due to time limits, you should aim for your presentation to **last 4 - 4.5 min** and it **should not exceed 5 min**. **Each student in a team must present**. You are expected to record your presentation with a video and submit it.

You will have to prepare your presentation slides and submit it.

Note: It is perfectly fine if you do not have all (or any) of your results ready. Make sure to describe the experiments you are planning on running and explain what you think your experiments will demonstrate.

6. New Data

You are expected to get extra image data from internet or take them by your own. These are treated as additional dataset to be processed, in addition to the existing dataset.

7. Project Grading Policy

The final project makes up 20% of the course grade.

a) Report Grading Rubric (/100)

- Abstract (/10)
 - Provides a concise summary of the project and effectively highlights key insights/findings.
- Introduction/Motivation & Related Work (/20)
 - Is the problem described clearly? Why is the problem important (if you solved it, who would care)?
 - How have people tackled this problem before and how is your project building upon and different from prior work?
 - What is your goal: What are you going to check / test / validate / analyze / improve?
- Methodology (/20)

- The hypothesis is stated clearly: what are you trying to show / investigate?
- The dataset is described clearly.
- The method is described clearly (preprocessing steps, the model etc.).
- Performance metrics are meaningful and clearly defined.
- Experimental Results & Discussion (/25)
 - Key findings and insights of the project are effectively demonstrated and discussed.
 - Readers can understand the purpose of each experiment and how it relates to the research problem the project tackles.
 - Discusses limitations of the proposed method and/or analysis.
 - Discusses how the results and insights gained relate to and/or improve upon previous work.
 - For 2-person teams, there are higher expectations for experimental results (i.e., baselines, ablations, etc.).
- Figures (/5)
 - If your project generated results that are visual or better to understand when plotted, you must have figures that show the results.
 - If not, you must have at least one figure that illustrates the problem or the method that you are implementing to tackle the problem.
 - Figures are clear.
 - Captions are clear and tell the reader what to take away from the figure.
- New Ideas & Contributions (/20)
 - Does the project go beyond re-implementing an existing method and add something that has not been done before? This could involve a new analysis of the model, a change to the model, or a new application.
 - The larger the team, the higher the expectation. For teams of 3, adding an ablation study / running a new analysis is sufficient. For teams of 4, we expect more, either analyzing the method along several different axes, or making & ablating a proposed change to the model.
 - For model changes, we expect that changes to the model are clearly motivated with a hypothesis of what they will accomplish and an analyses that indicates whether the improvement materialized.

b) Video Grading Rubric (/100)

Note: Since presentations are holistic, we won't assign percentages to the individual sub-points.

- Clarity (/40)
 - Was your video easy to follow? Was it clear to someone unfamiliar with the particulars of the project?
 - Were your slides clear, concise, and well organized? Did they get the message across?

- Was your talk supported by meaningful visuals with a clear message? Were they easy to understand and legible?
- Did you articulate your thoughts effectively? Did you speak clearly and at a good pace to ensure others can follow your talk?
- Was the delivery of video well organized? Did you cover all your slides within the time limit without rushing through them?
- Was the coordination between joint speakers handled smoothly? Was the presentation divided up effectively and the presentation time well balanced between joint speakers?
- Content (/60)
 - Intro/Motivation & Related work: What are you trying to do? How is it done, and what are the limits of current practice?
 - Methodology: What is your approach, how is it different from previous work, and why is it better?
 - Results & Discussion: How did you evaluate your approach or conduct your analysis? Which experiments did you run to support your claims? What other experiments will you run and why? What are the limitations of your approach or analysis? (As mentioned above, not having your results ready for the presentation is perfectly fine. In that case, please make sure to discuss the experiments you will run and what you think they will demonstrate.)

c) Topic Grading Rubric (/100)

- Novelty (/40)
 - Is it a novel question?
 - Have you proposed a good solution?
- Data (/10)
 - Have you collected any new data?
 - Have you did special data processing?
- Complexity (/50)
 - How complex is your network structure?
 - Is the problem being solved difficult?

d) Code Grading Rubric (/100)

- Algorithm (/70)
 - The algorithm is excellent for the problem to solve.
 - The algorithm is described clearly.
 - Algorithms are reproducible.
 - Algorithms are innovative
- Executable (/10)
 - This program is executable.
- Code Style (/10)

- A clear coding style should consistently be applied, greatly enhancing program readability
- Comments (/10)
 - Comments are appropriately added and explain the program very well.